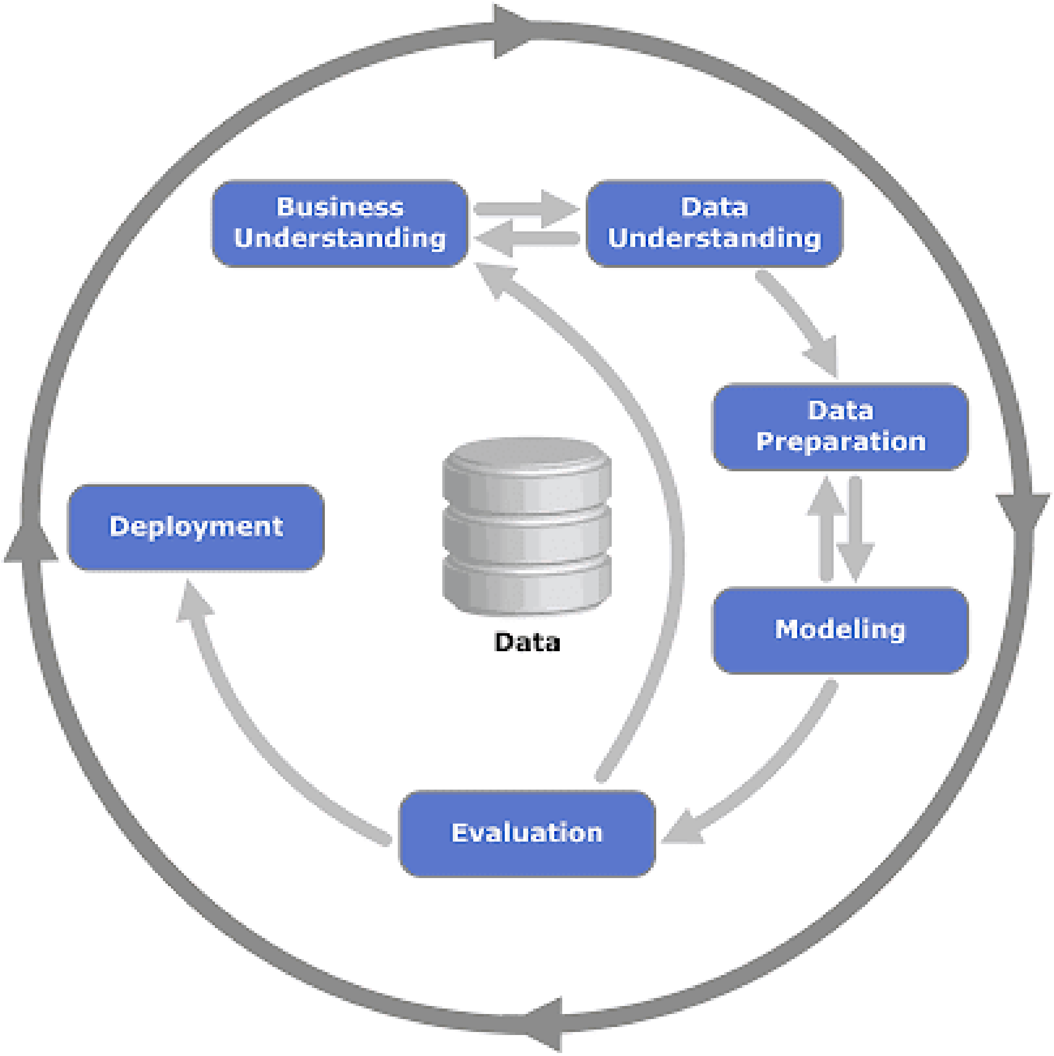
Data Modelling:

Regression and Classification

CRISP-DM


Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment
Data

Goals of Modelling

“Learned” by computer

Explanatory Relationship + Data = Prediction

Explanatory Relationship

1) Cause and effect
2) Dependencies between attributes

Data

Historical data serving as Experience

Prediction

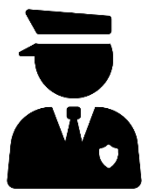
1. Based on past known behavior, what is going to happen in the future?
2. Provide interventions based on predicted outcomes (from reactive to proactive)

Machine learns via two main styles

tidy data format
col -> attribute
row -> obs

I. Supervised Learning:

- Process of an algorithm learning from the training dataset where it can be thought of as **a teacher supervising the learning process**.
- We provide the algorithm with the "correct answers", known as **labels/targets** during training.
- The algorithm "learn" how the rest of the features relate to the target, and make predictions about future outcomes.
- Majority of practical machine learning uses supervised learning.



This is a cat….
That is a dog…

Cover the other style next lecture

DOG

CAT

A simple example

- Identify the species of flower from the measurements of a flower.
- The data (the infamous ‘Iris Dataset’) is comprised of four flower measurements in cm (the columns of the data).
- Each row/observation of data is one example of an iris flower that has been measured and its known species.
- Can we tell what species an iris belongs to if we know its measurements?

	A	B	C	D	E
1	Sepal length	Sepal width	Petal length	Petal width	Species
2	5.1	3.5	1.4	0.2	I. setosa
3	4.9	3	1.4	0.2	I. setosa
4	4.7	3.2	1.3	0.2	I. setosa
5	4.6	3.1	1.5	0.2	I. setosa
6	5	3.6	1.4	0.2	I. setosa
7	5.4	3.9	1.7	0.4	I. setosa
8	4.6	3.4	1.4	0.3	I. setosa
9	5	3.4	1.5	0.2	I. setosa

Sample of Iris flower data

Target to predict

But need this label to train machine first

Simplest Predictive Modelling

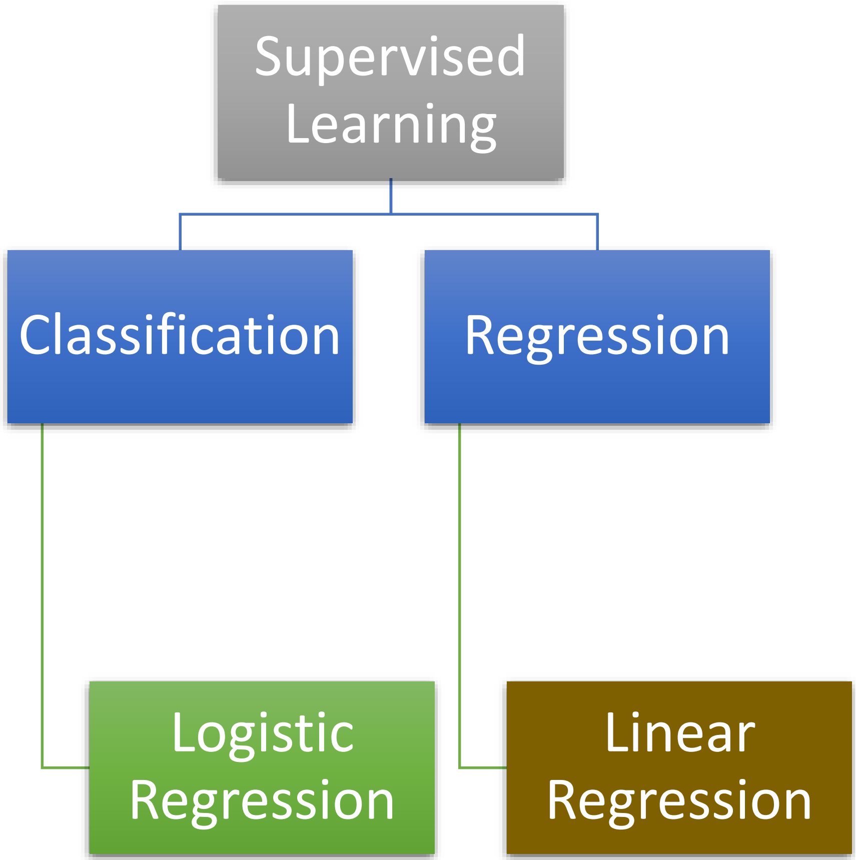
- Are you concerned about **which type** or **what is the value**?

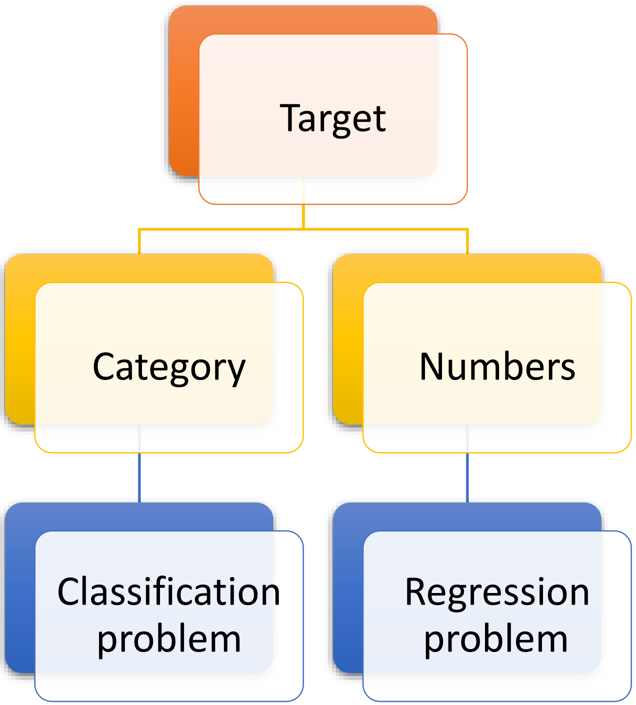
In the automobile manufacturing domain,
- the sales unit of vehicles is indirectly proportional to the price of the vehicles
- the price of a vehicle is decided by many factors such as usages of different metals/components, and various vehicle features such as RPM, mileage, length, width, and others.
- How can a manufacturer predict the ***amount of sales***?

In the banking domain, how can a bank predict the ***number of people*** applying for home loans, car loans, and personal loans, so that they can maintain their liquid capital to support the demand?

In the banking domain, prediction of default probability can be estimated using predictive modelling so that a bank can decide whether to ***approve/disapprove*** a loan to a customer.

Two types of typical problems solved with Supervised Learning





How to tell?

Linear Regression

- The linear regression model is used for explaining the relationship between a single dependent variable(Y) (known as the **target**) and one or more independent variables (X) known as inputs, predictors, features, attributes.

- Y must be a continuous variable
- X can be continuous or categorical

Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 \ldots\ldots \beta_n X_n + \varepsilon$$

- The linear regression model can be described using a general equation like this.
- The primary objective of a regression model is to estimate the beta parameters and minimize the error term epsilon.
- The values of beta parameters tell us the importance of the feature.
- If there is only ONE X, then this is a **simple** linear regression model.
- If there are more than one X, then this is a **multiple** linear regression model.

How to use the model

Linear Regression | Logistic Regression

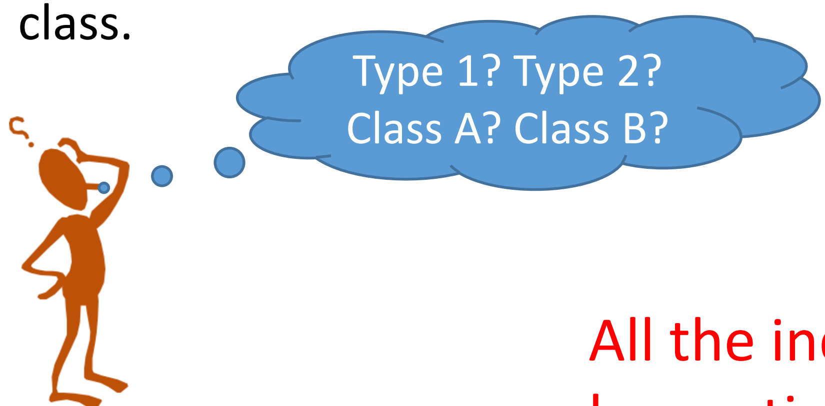
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 \ldots \beta_n X_n + \varepsilon$$

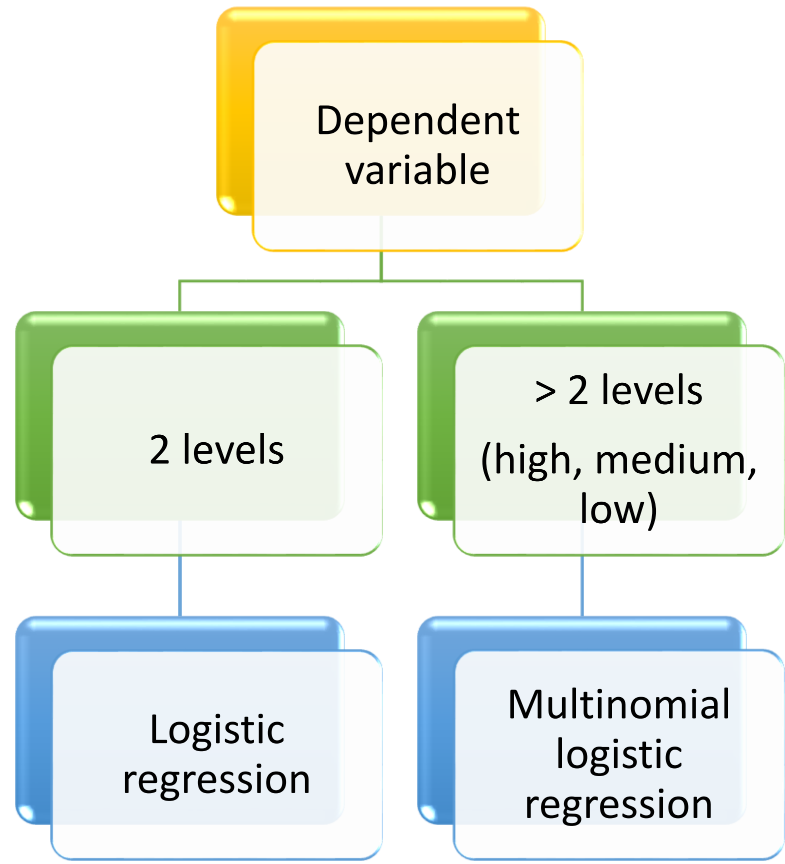
The regression analysis includes:

- Predicting the **future values** of the (dependent variable) target, given the values for attributes (independent variables)
- Assessment of **model fit** statistics (e.g. error rate, goodness of fit) and comparison of various models
- Interpreting the (beta parameters) coefficients to understand the **feature impact and relative importance** (e.g. a small change sway the outcome).

Logistic regression

- Real-life scenarios where the variable of interest is categorical in nature.
  - Should I buy a product ***or not***?
  - Has the credit card been approved?
  - Is the tumor cancerous ?
- Logistic regression **predicts** a dependent variable **class** AND also provides the **probability** of the observation belonging that predicted class.





All the independent variables/features (X) can be continuous, categorical, or nominal

Any more applications

Linear Regression

Logistic Regression

?

Linear vs Logistic regression model

<table>
<tr><th></th><th>Linear Regression</th><th>Logistic Regression</th></tr>
<tr><td rowspan="4">Outcome/ Dependent variable</td><td>Can have an infinite number of possible values.</td><td>Has only a limited number of possible values.</td></tr>
<tr><td>Continuous</td><td>Categorical</td></tr>
<tr><td colspan="2">With data relating to:
-> area in square feet of houses
-> corresponding sale price of those houses</td></tr>
<tr><td>• To predict selling price as a function of house size.
• Can have many possible outcome values</td><td>• To predict, based on size, whether a house would sell for more than $200K.
• The possible outputs are either Yes, the house will sell for more than $200K, or No, the house will not.</td></tr>
<tr><td>Co-efficient interpretation</td><td>Quite straightforward (i.e. holding all other variables constant, with a unit increase in this variable, the dependent variable is expected to increase/decrease by its beta value).</td><td>Depends on the distribution equation used (Binomial, Poisson, Multinomial, Ordinal, etc.) use, the interpretation is different. (need to substitute into equation to get the change)</td></tr>
<tr><td>Error minimization technique</td><td>Uses ordinary least squares (OLS) method to minimise the errors and arrive at a best possible fit</td><td>Uses maximum likelihood estimation (MLE) method to arrive at the solution</td></tr>
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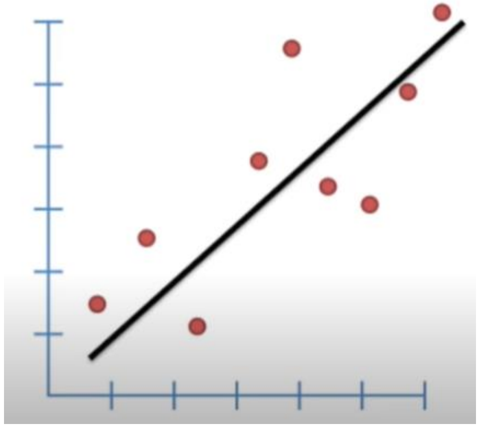
Additional Resources to OLS and MLE

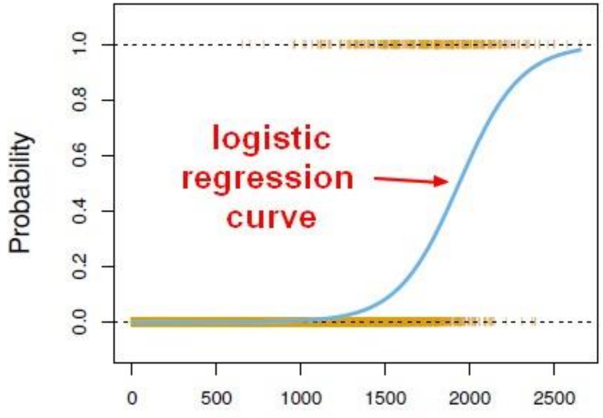
StatQuest: Fitting a line to data, aka least squares, aka linear regression.

https://youtu.be/PaFPbb66DxQ

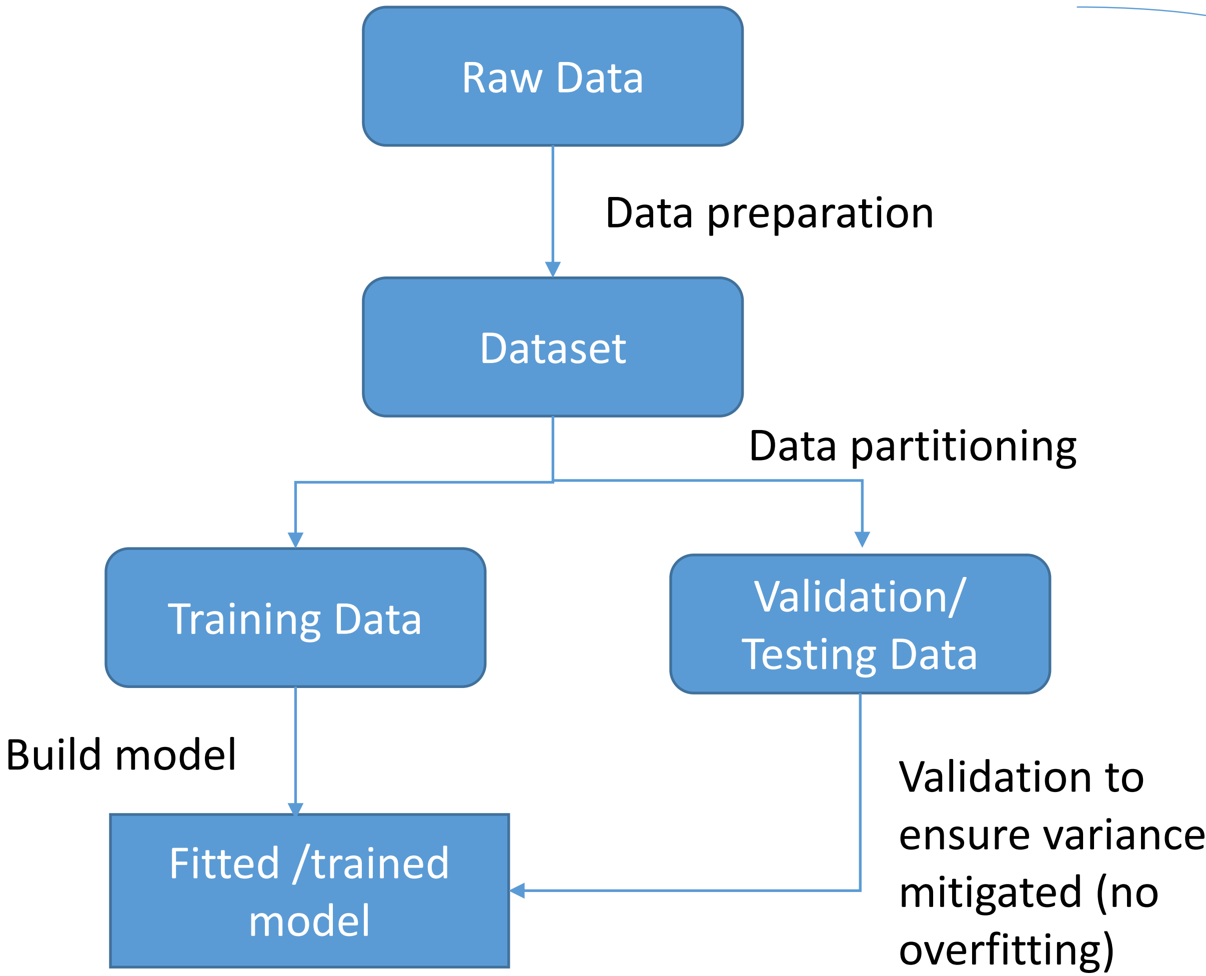


Logistic Regression Details: Maximum Likelihood

https://youtu.be/BfKanl1aSG0



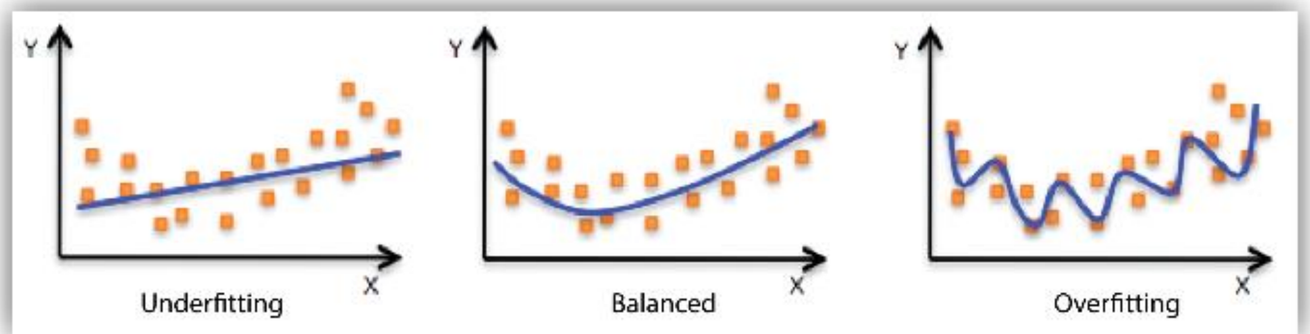
What is the proper sequence to build a model



Repeat to build other models using features prepared different and/or other relevant algorithms

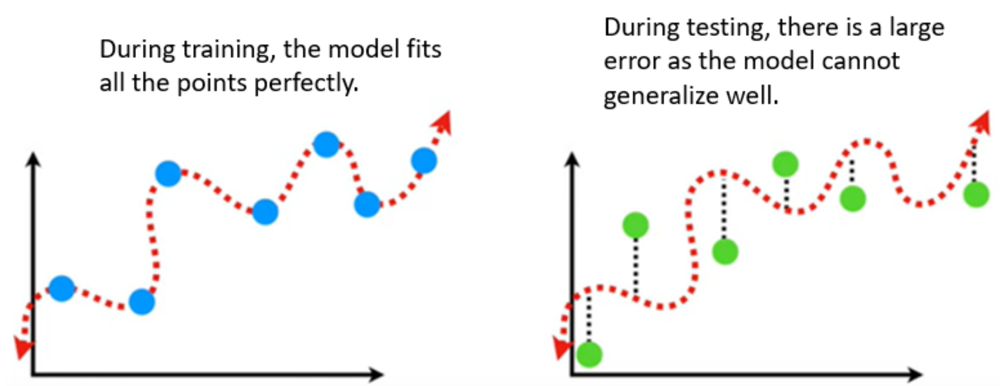
Select the best performer as the final model

Reduce Bias and Variance



An underfit (too simple) model barely pays attention to the training data by making **strong assumptions**, which makes it have high **bias**.

An overfit (too complicated) model tends to ‘**memorise**’ the training data points and does not perform well for test data. It has high **variance** (where variance refers to how much the model is dependent on data).



WE ❤

DATA

SCIENCE